Opgavesæt 1 i vektoranalyse

Alexander Nygaard Thomsen

1 Exercise 1

Let F be a vector field on $\mathbb{R}^3$ given by:

$$F(x,y,z) = (xy^2z, x^2yz, \frac{1}{2}x^2y^2)$$

calculate the integral:

$$\oint_C \mathbb{F} \cdot dr$$

where $C = \{(x,y,z) \in \mathbb{R}^3 | x^2 + y^2 = 1, z = 4\}$ (for positive orientation in the xy-plane)

Because x and y is defined as a unit circle in C, with a constant Z value, this can be rewritte as using polar coordinates, which becomes

$$x = cos(\theta), \qquad y = sin(\theta), \qquad z = 4$$

This changes the bound of integration to:

$$\int_0^{2\pi}$$

$$r(\theta) = \begin{pmatrix} cos(\theta) \\ sin(\theta) \\ 4 \end{pmatrix}$$

And F can also be rewritte in terms of parameter θ as **dr**, where:

$$dr(\theta) = \begin{pmatrix} -sin(\theta) \\ cos(\theta) \\ 0 \end{pmatrix}$$

which makes $F(dr(\theta))$

$$F(dr(\theta)) = \begin{pmatrix} 4cos(\theta)sin^2(\theta) \\ 4sin(\theta)cos(\theta) \\ \frac{1}{2}cos^2(\theta)sin^2(\theta) \end{pmatrix}$$

This makes the line integral

$$\int_0^{2\pi} \mathbb{F} \cdot dr \qquad = \qquad \begin{pmatrix} 4cos(\theta)sin^2(\theta) \\ 4sin(\theta)cos(\theta) \\ \frac{1}{2}cos^2(\theta)sin^2(\theta) \end{pmatrix} \cdot \begin{pmatrix} -sin(\theta) \\ cos(\theta) \\ 0 \end{pmatrix}$$

Yielding two terms

$$\int_0^{2\pi} -(4cos(\theta)sin^3(\theta)) + (4cos^3(\theta)sin(\theta))d\theta$$

This will be split up into two integrals and then solved
First term

$$\int -(4cos(\theta)sin^3(\theta))d\theta$$

$$\frac{d}{d\theta}sin(\theta) = cos(\theta), \qquad u = sin(\theta), \qquad \frac{du}{d\theta} = cos(\theta), \qquad du = cos(\theta)$$

$$-4\int cos(\theta)d\theta sin^3(\theta) \Rightarrow -4\int u^3 du \Rightarrow -4\frac{u^4}{4} + C = -sin^4(\theta) + C$$

Second Term

$$\int (4cos^3(\theta)sin(\theta))d\theta$$

$$\frac{d}{d\theta}cos(\theta) = -sin(\theta), \qquad u = cos(\theta), \qquad du = -sin(\theta), \qquad sin(\theta)d\theta = -du$$

$$4\int cos^3(\theta)sin(\theta)d\theta \Rightarrow 4\int u^3(-du) \Rightarrow -4\frac{u^4}{4} + C = -cos^4(\theta) + C$$

$$\left[-sin^4(\theta) - cos^4(\theta)\right]_0^{2\pi} \Rightarrow (-sin^4(2\pi) - cos^4(2\pi)) - (-sin^4(0) - cos^4(0)) = (-0^4 - 1^4) - (-0^4 - 1^4) = (-1) - (-1) = 0$$

2 Exercise 2

Let F be a vector field on $\mathbb{R}^3$ given by:

$$F(x, y, z) = (3x, 2y, -z)$$

- *Calculate $\nabla \cdot F$ and $\nabla \times F$ and find all function ϕ such that $\nabla\phi = F$*
- *Calculate the flux of F out of the unit sphere $S^2 = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 + z^2 = 1\}$*

The product of ∇ and F, which is vector field, yields a scalar field, this can be found as:

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x}(3x) + \frac{\partial F_y}{\partial y}(2y) + \frac{\partial F_z}{\partial z}(-z) = 3 + 2 - 1 = 4$$

This is also known as the divergence, this tells us something about the flux of the vector field, leaving a small closed surface. The curl of the vector field tells us about local rotation and can calculated as follows:

$$\nabla \times \mathbf{F} = \begin{pmatrix} \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \\ \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \\ \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\partial(-z)}{\partial y} - \frac{\partial(2y)}{\partial z} \\ \frac{\partial(3x)}{\partial z} - \frac{\partial(-z)}{\partial x} \\ \frac{\partial(2y)}{\partial x} - \frac{\partial(3x)}{\partial y} \end{pmatrix} = (0 - 0, 0 - 0, 0 - 0) = 0$$

This tells us that the vector field is *Irrotational* and also convervative. This also means that the field is defined everywhere in the space. If a vector field $\mathbf{F}$ is convervative, then the corresponding scalar field ϕ is called the potential for $\mathbf{F}$. The gradient consists of three partial derivates that must yield the vector field itself:

$$\frac{\partial \phi}{\partial x} = 3x \qquad \frac{\partial \phi}{\partial y} = 2y \qquad \frac{\partial \phi}{\partial z} = -z$$

Integrate the first term with respect to x, however we need to introduce arbitrary functions g(y,z) and h(z) during each integration, to recover any lost information, even though in this particular case, the different terms are not coupled:

$$\int \frac{\partial 3x}{\partial x} dx \Rightarrow \phi = \frac{3x^2}{2} + g(y, z)$$

$$\int g(y, z)dy = 2\frac{1}{2}y^2 + h(z) = y^2 + h(z) \Rightarrow \phi = \frac{3x^2}{2} + y^2 + h(z)$$

$$\int h(z)dz = -1\frac{1}{2}z^2 = -\frac{z^2}{2} + C$$

$$\phi = \frac{3x^2}{2} + y^2 - \frac{z^2}{2} + C$$

To find the flux flowing out of the unit sphere, the divergence theorem can be used. as it describes the flux: Here we transform the surface S into a solid sphere or ball B, but this also requires a change to spherical coordinates: (r, θ, ϕ)

$$\iint_{S^2} \mathbf{F} \cdot ndS = \iiint_B \nabla \cdot \mathbf{F} dV$$

from ealier, the divergence was calculated:

$$\nabla \cdot \mathbf{F} = 3 + 2 - 1 = 4$$

This yields the integral:

$$\iiint_B 4dV$$

Where $dV = r^2 sin\phi d\theta d\phi dr$, where the bounds of integration are defined as:

$$0 \leq r \leq R, \qquad 0 \leq \phi \leq \pi, \qquad 0 \leq \theta \leq 2\pi$$

$$\Phi = \int_0^1 \int_0^\pi \int_0^{2\pi} 4r^2 sin\phi d\theta d\phi dr$$

$$= \int_0^{2\pi} 4r^2 sin\phi d\theta \Rightarrow 4r^2 sin\phi \int_0^{2\pi} 1 d\theta \Rightarrow 4r^2 sin\phi \left[\theta\right]_0^{2\pi} = 4r^2 sin\phi(2\pi - 0) = 8\pi r^2 sin\phi$$

$$= \int_0^\pi 8\pi r^2 sin\phi d\phi \Rightarrow = 8\pi r^2 \int_0^\pi sin\phi d\phi \Rightarrow 8\pi r^2 \left[-cos\right]_0^\pi$$

$$\Rightarrow 8\pi r^2((-cos(\pi)) - (-cos(0))) = 8\pi r^2((-(-1)) - (-1)) = 16\pi r^2$$

$$\Phi = \int_0^1 16\pi r^2 dr \Rightarrow 16\pi \int_0^1 r^2 dr \Rightarrow \left[\frac{16\pi}{3} r^3\right]_0^1 = \frac{16\pi}{3}$$

3 Exercise 3

Define the potential ϕ as:

$$\phi(x,y,z) = ln(x^2+y^2+z^2), where (x,y,z) \in \mathbb{R}^3\{(0,0,0)\}$$

- Calculate $\nabla\phi$
- Calculate the flux of $\nabla\phi$ out of the sphere: $S^2 = \{(x,y,z) \in \mathbb{R}^3 | x^2+y^2+z^2 = R^2\}$ where R is a positive real number.

The gradient is defined as the partial derivative of each of the of the variables. In this case the potential ϕ is is a logarithmic function, meaning it has to be treated as an inner and an outer function, using the chain rule. $f(x) = ln(g(x)) \Rightarrow \frac{d}{dx}f(x) \Rightarrow f'(x) = \frac{1}{g(x)} \cdot g(x)$

$$\frac{\partial\phi}{\partial x} = \frac{\partial}{\partial x} ln(x^2+y^2+z^2) = \frac{1}{x^2+y^2+z^2} \cdot (2x+0+0) = \frac{2x}{x^2+y^2+z^2}$$
$$\frac{\partial\phi}{\partial y} = \frac{\partial}{\partial y} ln(x^2+y^2+z^2) = \frac{1}{x^2+y^2+z^2} \cdot (0+2y+0) = \frac{2y}{x^2+y^2+z^2}$$
$$\frac{\partial\phi}{\partial z} = \frac{\partial}{\partial z} ln(x^2+y^2+z^2) = \frac{1}{x^2+y^2+z^2} \cdot (0+0+2z) = \frac{2z}{x^2+y^2+z^2}$$

Yielding the final result:

$$\nabla\phi = \left(\frac{2x}{x^2+y^2+z^2}, \frac{2y}{x^2+y^2+z^2}, \frac{2z}{x^2+y^2+z^2}\right)$$

To calculate the flux out of the unit sphere S^2, this yields another scalar field, which is $\nabla \cdot \nabla$, this is also known as the *laplacian* of ϕ. This is possible if the scalar field is twice differentiable, which this is. And then afterwards integrate over the volume enclosed by the sphere.

$$\nabla \cdot \nabla\phi = \nabla^2\phi = \frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial\phi}{\partial z}\right)$$

Deriving the laplacian:

$$\nabla^2\phi = \frac{\partial}{\partial x}\left(\frac{2x}{x^2+y^2+z^2}\right) + \frac{\partial}{\partial y}\left(\frac{2y}{x^2+y^2+z^2}\right) + \frac{\partial}{\partial z}\left(\frac{2z}{x^2+y^2+z^2}\right)$$

This is evalutated using the quotiend rule:$(\frac{u}{v})' = \frac{u'v-uv'}{v^2}$

$$\frac{\partial}{\partial x}\left(\frac{2x}{x^2+y^2+z^2}\right) = \frac{2(x^2+y^2+z^2)-4x^2}{(x^2+y^2+z^2)^2}$$

The same procedure is done to derive the partial derivation expression for y and x, finally yielding the expressiom

$$\nabla^2\phi = \frac{2(x^2+y^2+z^2)-4x^2-2(x^2+y^2+z^2)-4y^2+2(x^2+y^2+z^2)-4z^2}{(x^2+y^2+z^2)^2}$$

$$= \frac{6(x^2+y^2+z^2)-4(x^2+y^2+z^)}{(x^2+y^2+z^2)^2}$$

Cleaning up the expression:

$$= \frac{2}{x^2+y^2+z^2}$$

And by the definition of the sphere, this can be expressed as:

$$\nabla\phi = \frac{2(x^2+y^2+z^2)}{x^2+y^2+z^2} = \frac{2}{R^2}$$

Flux through the sphere can then be evalutated as:

$$\Phi = \iiint_{S^2} \frac{2}{R^2} dV$$

Using the same bounds of integration as before:

$$\Phi = \int_0^1 \int_0^\pi \int_0^{2\pi} 2sin\theta \; d\phi \; d\theta \; dr$$

first integral:

$$2sin\theta \int_0^{2\pi} d\phi = 2sin\theta \left[\phi\right]_0^{2\pi} = 2sin\theta(2\pi - 0) = 4\pi sin\theta$$

second integral:

$$\int_0^\pi 4\pi sin\theta \; d\theta = 4\pi \int_0^\pi sin\theta \; d\theta = 4\pi \left[-cos\theta\right]_0^\pi = 4\pi \; ((-cos\pi) - (-cos(0))) = 4\pi \; (-(-1) - (-1)) = 4\pi \; (1+1) = 8\pi$$

third integral:

$$\int_0^1 8\pi dr = 8\pi \int_0^1 dr = 8\pi \; \left[r\right]_0^1 = 8\pi \; (1-0) = 8\pi$$

This results in the flux being:

$$\Phi = 8\pi$$

4 Exercise 4

Let $C = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 - 2x = 3\}$, (positive orientation) and let S be the line integral:

$$S = \oint_C (x + y)dx + (y - x)dy$$

- Calculate the line integral S directly
- Calculate the line integral using Greens Theorem

To compute this integral, C can firstly be rearranged:

$$C = x^2 + y^2 - 2x = 3 \Rightarrow x^2 - 2x + y^2 = 3 \Rightarrow (x - 1)^2 + y^2 = 4$$

This is an equation for a cicle, as it takes the form $x^2 + y^2 = r^2$ with a center in $[1, 0]$ and $r = 2$. Because this is in the positive orientation, that means the rotation is counter-clockwise, therefore the standard parameterization is:

$$x^2 + y^2 = r^2 \qquad \Rightarrow \qquad x = r\ cos(t), \qquad y = r\ sin(t), \qquad 0 \leq t \leq 2\pi$$

These can now be differentiated so that everything can be expressed in term of t:

$$\frac{dt}{dx} = -r\ sin(t) \qquad \frac{dt}{dy} = r\ cos(t)$$

$$S = \oint_C (x + y)dx + (y - x)dy \Rightarrow \int_0^{2\pi} (1 + 2cos(t) + 2sin(t))(-2sin(t))dt + (2sin(t) - 1 + cos(t))(2cos(t))dt$$

This expression can be simplfied

$$(-2)sin(t) - 4sin(t)\ cos(t) - 4sin^2(t) + 4sin(t)\ cos(t) - 2cos(t) - 4cos^2(t)$$

$$(-2)sin(t) - 2cos(t) - 4(sin^2(t) + cos^2(t))$$

$$(-2)(sin(t) + cos(t) - 4)$$

$$\int_0^{2\pi} \left[(-2)(sin(t) + cos(t)) - 4\right] dt \Rightarrow (-2) \int sin(t)dt - 2 \int cos(t)dt - 4 \int dt$$

$$\left[2cos(t) - 2sin(t) - 4t\right]_0^{2\pi} \Rightarrow (2cos(2\pi) - 2sin(2\pi) - 4(2\pi)) - (2cos(0) - 2sin(0) - 4(0))$$

$$= 2 \cdot 1 - 0 - 8\pi - 2 \cdot 1 - 0 - 0$$

$$= -8\pi$$

This can also be evalutated using Greens Theorem:

$$\oint_C Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Here, here functions Q and P must firstly be identified from the given contour integral:

$$P = (x + y) \qquad Q = (y - x), \qquad \frac{\partial Q}{\partial x} = -1 \qquad \frac{\partial P}{\partial y} = 1$$

Into green's theorem:

$$\oint_C (x + y)dx + (y - x)dy = \iint_D (-1 - 1)\ dA \Rightarrow \iint_D -2\ dA$$

The same bounds of integration can be derived by shifting into polar coordinates, and since this is still a circle it takes the same form as before:

$$x = r\ cos(\theta), \qquad y = r\ sin(\theta),$$

And the bounds:

$$0 \leq r \leq 2, \qquad 0 \leq \theta \leq 2\pi$$

And:

$$dA = r \; dr \; d\theta$$

$$S = \int_0^{2\pi} \int_0^2 -2r \; dr \; d\theta$$

$$-2 \int_0^2 r \; dr \Rightarrow -2r^2 \Rightarrow \left[-2r^2\right]_0^2 = (-2^2) - (-0^2) = -4$$

$$-4 \int_0^{2\pi} d\theta \Rightarrow -4\theta \Rightarrow [-4\theta]_0^{2\pi} = (-4 \cdot 2\pi) - (-4 \cdot 0)$$

$$= -8\pi$$